

Xing Chen

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EDUCATION

Evanston, IL

Northwestern University

September 2022 - June 2026

Major: B.S. Mechanical Engineering

Courses: Fluid Mechanics, System Dynamics, Heat Transfer, Computational Methods in Engineering Design, Physics
Mechatronics, Vehicle Dynamics and Design, Mechanics and Science of Materials, Chemistry, and Engineering Analysis.

TECHNICAL PROJECTS & RESEARCH

Battery Drop Certification Tester

January 2026 – June 2026

- Designed and built a battery drop testing mechanism for UL Solutions enabling repeatable safety certification of small-to-medium battery cells across standardized drop conditions, replacing outdated manual testing mechanisms.
- Engineered four core subsystems: pneumatic piston battery release mechanism, universal pin end effectors, lead screw height adjustment assembly, and drop base containment and safety monitoring system.
- Innovated a drop orientation measurement method that captures full force mapping across battery cell surfaces on impact, replacing incomplete corner-only contact data and significantly improving overall test accuracy for safety validation.
- Led full prototype fabrication from ideation and material sourcing through manufacturing and assembly, and served as primary editor for all engineering analysis, technical documentation, CAD designs, and presentations.

Gear Shaft Design Project

September 2025 – December 2025

- Designed a gear shaft assembly that receives rotational input from an electric motor and evenly distributes torque across two independent output shafts, each delivering 50% of total load with minimal power loss under varying operating conditions.
- Executed full mechanical design of a three-shaft, four-element gear train with six bearing selections to meet load and deflection requirements, 2D engineering drawings, and detailed 3D CAD simulations in SolidWorks.

Jack-In-A-Box Dynamic Simulation

March 2025 – June 2025

- Developed a Python simulation modeling the full dynamic motion of a jack bouncing inside a box, incorporating elastic impact derivations for collisions between the jack's rotational leg elements and the rigid box interior walls.
- Implemented full rigid body dynamics to compute real-time position, linear velocity, and rotational state of the jack across all phases of motion, culminating in a fully interactive 2D visual simulation output rendered in Matplotlib.

Pressure Vessel Optimization

January 2025 – March 2025

- Developed a parametric pressure vessel assembly model in Abaqus for Finite Element Analysis, enabling fully automated geometry variation and mesh updates across all optimization trials to support rapid design iteration.
- Implemented a Python pipeline integrated with Abaqus to minimize the total mass of a nuclear pressure vessel using Bayesian Optimization, significantly reducing design search time over conventional methods.
- Validated optimization results through multi-trial convergence analysis and systematic benchmarking against alternative approaches including random sampling and grid search, confirming overall solution robustness and reliability.

Injection Molding

January 2025 – March 2025

- Designed core and cavity molds for axolotl and boat assemblies, then machined both to precise tolerances using a Haas CNC milling machine and injection molded final components using the Morgan Press.
- Performed a metrology study to evaluate manufacturing quality, validating part tolerances, dimensional accuracy, and surface finish consistency across produced components.

Robotic Egg Lifting Project

March 2024 – May 2024

- Designed a two-fingered nylon end effector in NX CAD for a motor-powered robotic hand, optimizing finger geometry for reliable and repeatable object grasping.
- Optimized the end effector to safely grip eggs and other delicate objects from the side without fracture, validating performance through iterative physical testing and design refinement.

RELEVANT EXPERIENCE

Convident LLC Startup

September 2024 – Current

- Designed a vinyl cover for a new vending machine and assisted with inventory management.

Northwestern Baja SAE Racing

March 2024 – June 2024

- Designed gears and shafts using NX Siemens and machined mechanical components for a dune buggy-style racing vehicle.
- Focused on gear junctions, load transfer, and vehicle maneuverability.

SKILLS

Computer: MATLAB, Python, C, NX Siemens (CAD, CAM), SolidWorks, Altair (Inspire), SIMULIA (Isight, Abaqus), Adobe

Laboratory: Spectroscopy Machine, Laser Diffraction Apparatus, Sound Detector, Electromagnetic Graphing, Metrology

Workshop: Milling, Soldering, Robotics (programming and construction), Circuits, Woodworking, CNC, Sand Casting, Sheet Metal Working, 3D printing, Sewing, Laser Cutting, Vinyl Printing

Language: Fluent Cantonese, Conversational Mandarin

Hobbies: League of Legends, Valorant, Tennis, Badminton, Soccer, Drawing, Flute, Video Editing/Content Creation